



ABHAY MAHAJAN

B.Tech. IIT Roorkee

- 11+ years teaching experience
- Mentored and taught around 20000+ students
- Produced 400+ IIT selections
- 10+ INMO selections
- 1 EGMO Selection



Enrol Now For the **MEGA VOLT**

Vedantu Olympiad Litmus Test For Class 5th – 12th

Absolutley Free!

Registration Link:

vdnt.in/MEGAVOLTYT

Appear for the test on

14th Aug, 2021

3:00 pm



Reasons why there is nothing like VOLT

- Get mentored by our experienced Master Teachers who have given many olympiad qualifiers over the years.
- Daily LIVE, Interactive classes & Home Assignments to master subject-wise concepts.
- Revise your concepts 24*7 with recorded live lectures of your favourite Master Teachers.
- Weekly Test prepared by subject experts to examine your learnings.
- Individual results analysis by Academic Mentors to review your performance.
- Get rewards for your hard work & win amazon vouchers up to INR 17500*.



Want to be the Next
Mathematical Olympiad Genius?

Here is your chance to make
Your Own Path to the Top & **Crack PRMO** with Ease



Register for

VEDANTU OLYMPIAD WIZARD

ALL INDIA PRMO MOCK TEST SERIES

& Appear for

PRMO MOCK TEST-4

Duration of the Exam - **3 hours**

Students of **Class 8th -12th**

Can Appear for the Test anytime in **24 hours**
on **15th Aug 2021, Sunday** from **5pm** onwards

Absolutely Free!!

Click the link to Register:

<https://vdnt.in/VOWPRERMO2021>



Join us on our
Social Media Handles
for daily updates, interesting
quizzes, fun facts and much more



[vdnt.in/VOSINSTA](https://www.instagram.com/vdnt.in/VOSINSTA)



[vdnt.in/VOSFACEBOOK](https://www.facebook.com/vdnt.in/VOSFACEBOOK)



[vdnt.in/volympiad](https://www.telegram.com/vdnt.in/volympiad)



[vdnt.in/VOSQUORA](https://www.quora.com/vdnt.in/VOSQUORA)



KVPY SA SCHEDULE – WEEK 2

**ABHAY
MAHAJAN**

Geometry- Triangles

**MONDAY
@ 11:00 AM**

**NISHANT
SHARMA**

**Work, Power, Energy &
Circular Motion**

**TUESDAY
@ 11:00 AM**

**ASIF
TYAGI**

Animal Kingdom

**WEDNESDAY
@ 4:00 PM**

**NIDHI
PRASHAR**

**Periodic Classification of
Elements & Chemical Bonding**

**SATURDAY
@ 11:30 AM**

KVPY SX SCHEDULE – WEEK 2

**ABHAY
MAHAJAN**

**Permutation and Combination and
Probability**

**WEDNESDAY
@ 11:00 AM**

**NISHANT
SHARMA**

Work, Power & Energy

**THURSDAY
@ 11:00 AM**

**ASIF
TYAGI**

**Animal Reproduction:
Part-2**

**FRIDAY
@ 4:00 PM**

**NIDHI
PRASHAR**

**Periodic Classification of
Elements & Chemical Bonding**

**SATURDAY
@ 11:30 AM**



Only on
VOS Youtube Channel

EVERY TUESDAY

PRE-RMO

4 PM

EVERY THURSDAY

PRE-RMO

4 PM

EVERY SATURDAY

PRE-RMO

4 PM

ABHAY MAHAJAN
IIT ROORKEE



For
DOUBTS , QUERIES & MOTIVATION
Reach out to me...



abhay86mahajan



abhay.mahajan@vedantu.com

ABHAY MAHAJAN
MATHEMATICS EXPERT

6



Content Creator :

Sahil Jindal

B.Tech, NIT Kurukshetra

4 Year JEE Teaching
Experience.



PRMO:

Cauchy Schwarz and Chebyshev's Inequality



Cauchy Schwarz Inequality

If $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ are any real numbers (+ve, zero or -ve)

$$\begin{array}{ccccccc} a_1 & a_2 & a_3 & \dots & a_n \\ b_1 & b_2 & b_3 & \dots & b_n \end{array}$$

$$\underline{(a_1b_1 + a_2b_2 + \dots + a_nb_n)^2 \leq (a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2)}$$

Equality hold, iff $a_1 : a_2 : \dots : a_n :: b_1 : b_2 : \dots : b_n$

$$\boxed{(\sum a_i b_i)^2 \leq \sum a_i^2 \cdot \sum b_i^2} \quad \checkmark$$

$$(a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) \quad ||$$

Cauchy Schwarz Inequality

$$(a_1b_1 + a_2b_2 + \dots + a_nb_n)^2 \leq (a_1^2 + a_2^2 + \dots + a_n^2) (b_1^2 + b_2^2 + \dots + b_n^2)$$

Proof : Consider a Quadratic

$$(a_1x + b_1)^2 + (a_2x + b_2)^2 + \dots + (a_nx + b_n)^2$$

$$= \sum_{i=1}^n a_i^2 x^2 + \sum_{i=1}^n b_i^2 + 2 \sum_{i=1}^n a_i b_i x \geq 0$$

Therefore this quadratic has at most one real root $\Rightarrow D \leq 0$

$$\sum_{i=1}^n a_i^2 \cdot \sum_{i=1}^n b_i^2 \geq \left(\sum_{i=1}^n a_i b_i \right)^2$$

Application:

$$a^2 + b^2 \geq \frac{(a+b)^2}{2}$$

✓
obvious

AM $\geq$ nth power

$$\frac{a^2 + b^2}{2} \geq \left(\frac{a+b}{2}\right)^2$$

✓
✓
✓

a, b

$1, 1$

$$a^2 + b^2 \geq \frac{(a+b)^2}{2}$$

$$(a^2 + b^2)(1^2 + 1^2) \geq (a+b)^2$$

$$\frac{(a+b)^2}{2} \leq a^2 + b^2$$

Corollary

Let $a_1, a_2, \dots, a_n; b_1, b_2, \dots, b_n$ be real numbers such that $b_1, b_2, \dots, b_n > 0$.

Then

$$\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \dots + \frac{a_n^2}{b_n} \geq \frac{(a_1 + a_2 + \dots + a_n)^2}{b_1 + b_2 + \dots + b_n},$$

with equality if and only if

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \dots = \frac{a_n}{b_n}.$$

Cute

$$\left(\frac{a_1}{b_1} + \frac{a_2}{b_2} + \dots + \frac{a_n}{b_n} \right)^2 \leq \left(\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \dots + \frac{a_n^2}{b_n} \right) (b_1 + b_2 + \dots + b_n)$$

Cauchy's Lemma

~~$\frac{2}{3} + \frac{4}{5} = \frac{6}{8}$~~

~~$\frac{3}{4} = \frac{3+1}{4+1} = \frac{4}{5}$~~

Chebyshev's Inequality

Let $a_1 \leq a_2 \leq \dots \leq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$ be real numbers. Then We have

$$\left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) \leq n \sum_{i=1}^n a_i b_i$$

$$(a_1 + a_2 + \dots + a_n) (b_1 + b_2 + \dots + b_n) \leq n (a_1 b_1 + a_2 b_2 + \dots + a_n b_n)$$

Equality occurs if and only if

$$a_1 = a_2 = \dots = a_n \text{ or } b_1 = b_2 = \dots = b_n$$

Ascending order

(TCV)

$$\leq a_i \leq b_i \leq n \leq a_n b_n$$

$$(a_1 + a_2 + a_3)(b_1 + b_2 + b_3) \leq 3(a_1 b_1 + a_2 b_2 + a_3 b_3)$$

$$(\leq a_i b_i)^2 \leq \leq a_i^2 \leq b_i^2$$

C.S.

$$(\sum a_i b_i)^2 \leq \sum a_i^2 \cdot \sum b_i^2$$

Q1. If $x^2 + y^2 + z^2 = 5$, Find the maximum value of $(x + 2y + 3z)^2$.

Direct Application

$$\textcircled{70} \quad (x + 2y + 3z)^2 \leq (1^2 + 2^2 + 3^2)(x^2 + y^2 + z^2)$$

$$k^2 + 4k^2 + 9k^2 = 5$$

$$k^2 = 5/14$$

$$\textcircled{k = \sqrt{5/14}}$$

$$= 14 \cdot 5$$

$$= \textcircled{70}$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{3} = k$$

$$x = \sqrt{\frac{5}{14}}, y = 2\sqrt{\frac{5}{14}}, z = 3\sqrt{\frac{5}{14}}$$

$$x = k, y = 2k, z = 3k$$

Q1. If $x^2 + y^2 + z^2 = 5$, Find the maximum value of $(x + 2y + 3z)^2$.

Answer: 70

Solution:

$$(x + 2y + 3z)^2 \leq (x^2 + y^2 + z^2)(1^2 + 2^2 + 3^2)$$

$$(x + 2y + 3z)^2 \leq 70$$

Hint 1

AMC 12 2013

$a, b \rightarrow$ nahi
Chahithe

Q2. Let a, b and c be real numbers such that

$$a + b + c = 2, \text{ and } a^2 + b^2 + c^2 = 12$$

What is the difference between the maximum and minimum possible values of c ?

$$\begin{array}{c} + \quad - \quad + \\ \hline -2 \quad \quad 10/3 \end{array}$$

A

2

B

$\frac{10}{3}$

C

~~4~~

D

$\frac{16}{3}$

$$a^2 + b^2 \geq \left(\frac{a+b}{2} \right)^2$$

$$12 - c^2 \geq \left(\frac{2-c}{2} \right)^2$$

$$24 - 2c^2 \geq 4 + c^2 - 4c$$

Application of CS

$$3c^2 - 4c - 20 \leq 0$$

$$3c^2 + 6c - 10c - 20 \leq 0$$

$$3c(c+2) - 10(c+2) \leq 0$$

$$(3c-10)(c+2) \leq 0$$

$$-2 \leq c \leq \frac{10}{3}$$

$$\frac{10}{3} - (-2)$$

Q2. Let a , b and c be real numbers such that

$$a + b + c = 2, \text{ and } a^2 + b^2 + c^2 = 12$$

What is the difference between the maximum and minimum possible values of c ?

A 2

B $\frac{10}{3}$

C 4

D $\frac{16}{3}$

Solution:

$a + b = 2 - c$. Now, by Cauchy- Schwarz, we have that

$$(a^2 + b^2) \geq \frac{(2-c)^2}{2}.$$

Therefore, we have that $\frac{(2-c)^2}{2} + c^2 \leq 12$.

We then find the roots of c that satisfy equality and find the difference of the roots. This gives the answer, $\frac{16}{3}$

Thru's $\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \frac{a_3^2}{b_3} \geq \frac{(a_1+a_2+a_3)^2}{b_1+b_2+b_3}$

PRMO - 2018

Q3. What is the largest positive integer n such that

$$\frac{\frac{a^2}{\frac{b}{29} + \frac{c}{31}}}{\frac{b}{29} + \frac{c}{31}} + \frac{\frac{b^2}{\frac{c}{29} + \frac{a}{31}}}{\frac{c}{29} + \frac{a}{31}} + \frac{\frac{c^2}{\frac{a}{29} + \frac{b}{31}}}{\frac{a}{29} + \frac{b}{31}} \geq n(a+b+c)$$

holds for all positive real numbers a, b, c .

A 11

B 12

C 14

D 15

$$\frac{a^2}{\frac{b}{29} + \frac{c}{31}} + \frac{b^2}{\frac{c}{29} + \frac{a}{31}} + \frac{c^2}{\frac{a}{29} + \frac{b}{31}} \geq \frac{(a+b+c)^2}{b(\frac{1}{29} + \frac{1}{31}) + a(\frac{1}{29} + \frac{1}{31}) + c(\frac{1}{29} + \frac{1}{31})}$$

$$\geq 14.9(a+b+c)$$

$$\text{K1} \geq 14(a+b+c)$$

$$\frac{(a+b+c)^2}{b(\frac{1}{29} + \frac{1}{31}) + a(\frac{1}{29} + \frac{1}{31}) + c(\frac{1}{29} + \frac{1}{31})} = \frac{(a+b+c)^2}{\frac{(31+29)(a+b+c)}{29 \cdot 31}} = \frac{29 \cdot 31}{60} (a+b+c)$$

$$\frac{29 \cdot 31}{60}$$

$$\frac{(30-1)(30+1)}{60}$$

$$= \frac{89.9}{6} = 14.9$$

$$a=b=c$$

$$n \leq 14.9$$

$$(a+b+c)$$

Q3. What is the largest positive integer n such that

$$\frac{a^2}{\frac{b}{29} + \frac{c}{31}} + \frac{b^2}{\frac{c}{29} + \frac{a}{31}} + \frac{c^2}{\frac{a}{29} + \frac{b}{31}} \geq n(a + b + c)$$

holds for all positive real numbers a, b, c .

A 11

B 12

C 14

D 15

Solution:

$$\begin{aligned} \frac{a^2}{x} + \frac{b^2}{y} + \frac{c^2}{z} &\geq \frac{(a+b+c)^2}{x+y+z} \\ \frac{a^2}{\frac{b}{29} + \frac{c}{31}} + \frac{b^2}{\frac{c}{29} + \frac{a}{31}} + \frac{c^2}{\frac{a}{29} + \frac{b}{31}} &\geq \frac{(a+b+c)^2}{a\left(\frac{1}{29} + \frac{1}{31}\right) + b\left(\frac{1}{29} + \frac{1}{31}\right) + c\left(\frac{1}{29} + \frac{1}{31}\right)} \\ &\geq \left(\frac{a+b+c}{\frac{1}{29} + \frac{1}{31}} \right) \\ &\geq \frac{a+b+c}{\frac{60}{29 \times 31}} \\ &\geq \frac{29 \times 31}{60} (a+b+c) \\ &\geq 14.98(a+b+c) \end{aligned}$$

So, $n = 14$

Q4. Let $a, b, c \in \mathbb{R}^+$. Prove the inequality

$$\frac{a^2}{3^3} + \frac{b^2}{4^3} + \frac{c^2}{5^3} \geq \frac{(a+b+c)^2}{6^3} \quad \checkmark \checkmark$$

$$3^3 + 4^3 + 5^3 = 6^3$$

$$\begin{array}{r} 27 + 64 + 125 \\ \hline = 216 \end{array}$$

$$\frac{a^2}{3^3} + \frac{b^2}{4^3} + \frac{c^2}{5^3} \geq \frac{(a+b+c)^2}{\underbrace{3^3 + 4^3 + 5^3}_{= 6^3}}$$

Direct application of Titu

Solution:

Note that $3^3 + 4^3 + 5^3 = 6^3$.

Taking

$$a_1 = \frac{a}{\sqrt{3^3}}, \quad a_2 = \frac{b}{\sqrt{4^3}}, \quad a_3 = \frac{c}{\sqrt{5^3}};$$

$$b_1 = \sqrt{3^3}, \quad b_2 = \sqrt{4^3}, \quad b_3 = \sqrt{5^3},$$

by the Cauchy - Schwarz inequality we obtain

$$\left(\frac{a^2}{3^3} + \frac{b^2}{4^3} + \frac{c^2}{5^3} \right) (3^3 + 4^3 + 5^3) \geq (a + b + c)^2,$$

as required.

$$\left[(\sqrt{b+c})^2 + (\sqrt{c+a})^2 + (\sqrt{a+b})^2 \right] \left[\left(\frac{3}{b+c} \right)^2 + \left(\frac{4}{c+a} \right)^2 + \left(\frac{5}{a+b} \right)^2 \right] \geq (\sqrt{3} + \sqrt{4} + \sqrt{5})^2$$

Q5. Let a, b, c be positive real numbers. Determine the minimal value of $\frac{3a}{b+c} + \frac{4b}{c+a} + \frac{5c}{a+b}$. = S

$$S = \frac{3a}{b+c} + 3 + \frac{4b}{c+a} + 4 + \frac{5c}{a+b} + 5 - 12$$

$$= 3 \frac{(a+b+c)}{b+c} + 4 \frac{(a+b+c)}{c+a} + 5 \frac{(a+b+c)}{a+b} - 12$$

$$= \frac{1}{2} (a+b+c) \left[\frac{3}{b+c} + \frac{4}{c+a} + \frac{5}{a+b} \right] - 12$$

$$S_{\min} = \frac{1}{2} \left[\sqrt{3} + \sqrt{4} + \sqrt{5} \right]^2 - 12$$

Q5. Let a, b, c be positive real numbers. Determine the minimal value of $\frac{3a}{b+c} + \frac{4b}{c+a} + \frac{5c}{a+b}$.

Answer: $\frac{1}{2}(\sqrt{3} + \sqrt{4} + \sqrt{5})^2 - 12$

Solution:

By the Cauchy- Schwarz inequality we have

$$\begin{aligned}& \frac{3a}{b+c} + \frac{4b}{c+a} + \frac{5c}{a+b} + (3+4+5) \\&= (a+b+c) \left(\frac{3}{b+c} + \frac{4}{c+a} + \frac{5}{a+b} \right) \\&= \frac{1}{2} ((b+c) + (c+a) + (a+b)) \left(\frac{3}{b+c} + \frac{4}{c+a} + \frac{5}{a+b} \right) \\&\geq \frac{1}{2} (\sqrt{3} + \sqrt{4} + \sqrt{5})^2.\end{aligned}$$

$$\text{Hence } \frac{3a}{b+c} + \frac{4b}{c+a} + \frac{5c}{a+b} \geq \frac{1}{2} (\sqrt{3} + \sqrt{4} + \sqrt{5})^2 - 12.$$

So the minimal value of the expression is $\frac{1}{2} (\sqrt{3} + \sqrt{4} + \sqrt{5})^2 - 12$,

and it is reached if and only if $\frac{b+c}{\sqrt{3}} = \frac{c+a}{2} = \frac{a+b}{\sqrt{5}}$

Cauchy Schwarz

JAPAN - 2010

$$a_1b_1 + a_2b_2 + a_3b_3$$

Q6. Let x, y, z be positive real numbers. Prove that

$$\frac{1+xy+xz}{(1+y+z)^2} + \frac{1+yz+yx}{(1+z+x)^2} + \frac{1+zx+zy}{(1+x+y)^2} \geq 1.$$

$$\geq \frac{y}{x+y+z}$$

$$\checkmark (a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

$$\frac{x+y+z}{x}$$

$$\frac{z}{x+y+z}$$

$$\begin{aligned} (1+y+z)^2 &\leq (1+xy+xz) \left(1+\frac{y}{x}+\frac{z}{x}\right) \\ (a_1b_1 + a_2b_2 + a_3b_3)^2 &\leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) \end{aligned}$$

$$\boxed{\frac{1+xy+xz}{(1+y+z)^2} \geq \frac{x}{x+y+z}} \quad \checkmark$$

Solution:

The Cauchy- Schwarz inequality yields

$$\left(1 + \frac{y}{x} + \frac{z}{x}\right)(1 + xy + xz) \geq (1 + y + z)^2$$

which implies $\frac{1 + xy + xz}{(1 + y + z)^2} \geq \frac{x}{x + y + z}.$

Similarly

$$\frac{1 + yz + yx}{(1 + z + x)^2} \geq \frac{y}{x + y + z}, \quad \frac{1 + zx + zy}{(1 + x + y)^2} \geq \frac{z}{x + y + z}.$$

By adding up these three inequalities, the desired inequality is obtained.



For
DOUBTS , QUERIES & MOTIVATION
Reach out to me...



abhay86mahajan



abhay.mahajan@vedantu.com

ABHAY MAHAJAN
MATHEMATICS EXPERT

APRIL
MONTH SCHEDULE



ABHAY MAHAJAN
MATHEMATICS EXPERT

Only on
VOS Youtube Channel



EVERY TUESDAY

NTSE MATHS

4 PM

EVERY THURSDAY

PRE-RMO

4 PM

EVERY SATURDAY

PRE-RMO

4 PM



KVPY SA SCHEDULE – WEEK 2

**ABHAY
MAHAJAN**

Geometry- Triangles

**MONDAY
@ 11:00 AM**

**NISHANT
SHARMA**

**Work, Power, Energy &
Circular Motion**

**TUESDAY
@ 11:00 AM**

**ASIF
TYAGI**

Animal Kingdom

**WEDNESDAY
@ 4:00 PM**

**NIDHI
PRASHAR**

**Periodic Classification of
Elements & Chemical Bonding**

**SATURDAY
@ 11:30 AM**

KVPY SX SCHEDULE – WEEK 2

**ABHAY
MAHAJAN**

**Permutation and Combination and
Probability**

**WEDNESDAY
@ 11:00 AM**

**NISHANT
SHARMA**

Work, Power & Energy

**THURSDAY
@ 11:00 AM**

**ASIF
TYAGI**

**Animal Reproduction:
Part-2**

**FRIDAY
@ 4:00 PM**

**NIDHI
PRASHAR**

**Periodic Classification of
Elements & Chemical Bonding**

**SATURDAY
@ 11:30 AM**

Join us on our
Social Media Handles
for daily updates, interesting
quizzes, fun facts and much more



[vdnt.in/VOSINSTA](https://www.instagram.com/vedantuolympiad...)



[vdnt.in/VOSFACEBOOK](https://www.facebook.com/vedantuolympiad...)



[vdnt.in/volympiad](https://www.telegram.com/vedantuolympiad...)



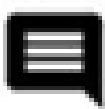
[vdnt.in/VOSQUORA](https://www.quora.com/vedantuolympiad...)



Let's Bring
GOLD
For
INDIA



If you liked the session, then

 **LIKE**
 **COMMENT**
 **SUBSCRIBE**
 **SHARE**

